

XR-2041 Phase III Randomised Controlled Trial

A Multicentre, Double-Blind, Placebo-Controlled Study of XR-2041
in Patients with Moderate-to-Severe Rheumatoid Arthritis

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Abstract

Background: Rheumatoid arthritis (RA) affects approximately 1% of the global population and is characterised by synovial inflammation and joint destruction. Current biologics achieve remission in only 30-40% of patients, leaving a significant unmet need for novel therapies.

Objective: To evaluate the efficacy and safety of XR-2041, a selective JAK1/JAK3 inhibitor, compared with placebo in adults with inadequate response to MTX.

Design: 52-week, Phase III, randomised (2:1), double-blind, placebo-controlled trial conducted across 48 sites in 12 countries.

Participants: 1,248 adults (18-75 yr) with ACR/EULAR-defined moderate-to-severe RA, DAS28-CRP ≥ 3.2 , and documented inadequate response to methotrexate (MTX).

Interventions: XR-2041 15 mg once daily (n=832) or matched placebo (n=416), both with stable background MTX 7.5-25 mg/week.

Primary Endpoint: ACR20 response rate at Week 12.

Key Secondary Endpoints: DAS28-CRP remission, ACR50/70 at Week 12 and 52, HAQ-DI change from baseline, radiographic progression (mTSS) at Week 52.

Results: ACR20 at Week 12: XR-2041 70.4% vs placebo 29.1% ($p < 0.001$, OR 5.82).
DAS28-CRP remission at Week 12: 38.2% vs 8.4% ($p < 0.001$).
ACR50 at Week 12: 48.7% vs 14.1%. ACR70 at Week 12: 26.3% vs 5.3%.
No new safety signals; serious adverse events: 6.1% XR-2041 vs 5.8% placebo.

Conclusion: XR-2041 demonstrated superior efficacy vs placebo across all primary and key secondary endpoints with an acceptable safety profile.

1. Introduction

Rheumatoid arthritis (RA) is a systemic autoimmune disease primarily affecting synovial joints, with a worldwide prevalence of 0.5-1.0%. The pathogenesis involves dysregulated cytokine signalling – particularly IL-6, TNF-alpha, and IL-17 – mediated through Janus kinase (JAK) pathways.

Despite the availability of conventional DMARDs and biologic agents (TNFi, IL-6Ri, anti-CD20), approximately 30-40% of patients fail to achieve sustained remission. JAK inhibitors represent the latest class of targeted synthetic DMARDs (tsDMARDs), with tofacitinib, baricitinib, and upadacitinib approved for RA.

XR-2041 is a next-generation oral JAK1/JAK3 inhibitor with high selectivity (JAK1 IC₅₀ = 3.2 nM; JAK3 IC₅₀ = 5.8 nM; >200-fold selectivity over JAK2), designed to minimise off-target haematological effects associated with JAK2 inhibition observed with earlier agents.

2. Methods

2.1 Study Design

This was a global, multicentre, randomised, double-blind, placebo-controlled, parallel-group Phase III study. Patients were randomised 2:1 to XR-2041 or placebo using interactive response technology (IRT), stratified by:

- Geographic region (North America / Europe / Asia-Pacific / Other)
- Prior biologic use (naïve vs experienced)
- Baseline DAS28-CRP (<5.1 vs ≥5.1)

2.2 Participants

Inclusion: Age 18-75; ACR/EULAR 2010 RA classification criteria; DAS28-CRP ≥3.2; stable MTX (≥12 weeks, 7.5-25 mg/week); inadequate response defined as ≥3 months of therapy without remission.

Exclusion: Prior JAK inhibitor use; active or latent TB (QuantiFERON positive); eGFR <30 mL/min/1.73m²; ALT/AST >2x ULN; current malignancy; pregnancy.

2.3 Interventions

XR-2041 15 mg film-coated tablet, once daily, orally.

Placebo: matched tablet, once daily, orally.

Background: stable MTX throughout; folic acid 5 mg/week mandatory.

Rescue: IV methylprednisolone 125 mg single dose permitted once for flare.

3. Results

3.1 Patient Disposition and Baseline

Screened: 1,847 | Randomised: 1,248 | Completed 52 weeks: 1,089 (87.3%)
XR-2041: n=832 (completed=731, 87.9%) | Placebo: n=416 (completed=358, 86.1%)

Baseline characteristics were well balanced:

Mean age: 51.3 yr (XR) vs 51.8 yr (PBO)

Female: 78.1% vs 77.6%

Mean DAS28-CRP: 5.81 vs 5.79

Mean HAQ-DI: 1.48 vs 1.51

Biologic-naïve: 71.4% vs 70.9%

Median RA duration: 7.2 yr vs 7.5 yr

Mean CRP: 18.4 mg/L vs 19.1 mg/L

Mean SJC28: 12.3 vs 12.6

3.2 Primary Endpoint

ACR20 Response at Week 12:

XR-2041: 586/832 (70.4%) vs Placebo: 121/416 (29.1%)

Odds Ratio: 5.82 (95% CI 4.31-7.86) | p-value: <0.0001

Risk Difference: 41.3% (95% CI 35.9-46.7%)

NNT: 2.4 (95% CI 2.1-2.8)

3.3 Key Secondary Endpoints at Week 12

DAS28-CRP <2.6 (remission): 38.2% vs 8.4% (OR 6.65; p<0.0001)

DAS28-CRP ≤3.2 (LDA): 56.7% vs 17.8% (OR 5.97; p<0.0001)

ACR50 response: 48.7% vs 14.1% (OR 5.67; p<0.0001)

ACR70 response: 26.3% vs 5.3% (OR 6.31; p<0.0001)

CDAI remission (≤2.8): 22.1% vs 4.1% (OR 6.60; p<0.0001)

HAQ-DI change from BL: -0.61 vs -0.22 (diff -0.39; p<0.0001)

Patient Global VAS change: -31.4 vs -12.8 (diff -18.6; p<0.0001)

3.4 Radiographic Outcomes at Week 52

mTSS change: 0.42 (XR) vs 1.87 (PBO) [diff -1.45; 95%CI -2.01/-0.89; p<0.0001]

Erosion score change: 0.18 vs 0.94 (diff -0.76; p<0.0001)

JSN score change: 0.24 vs 0.93 (diff -0.69; p<0.0001)

No radiographic progression (deltaTSS≤0): 68.4% vs 44.7% (p<0.0001)

3.5 Safety Summary

Treatment-emergent AEs (TEAEs): 74.3% XR-2041 vs 66.8% placebo

Serious TEAEs: 6.1% vs 5.8% | TEAE leading to discontinuation: 3.2% vs 2.4%

Most common TEAEs ($\geq 5\%$ in either arm):

Upper respiratory tract infection: 14.8% vs 10.6%

Nasopharyngitis: 9.2% vs 7.9%

Headache: 8.1% vs 6.7%

Nausea: 6.4% vs 3.8%

Diarrhoea: 5.3% vs 4.1%

Urinary tract infection: 5.1% vs 3.6%

Events of special interest:

Serious infections: 2.4% vs 2.2% (incidence rate 2.8 vs 2.5/100 PY)

Herpes zoster: 1.8% vs 0.7% (IR 2.1 vs 0.8/100 PY)

MACE: 0.2% vs 0.0% (IR 0.24/100 PY; adjudicated by independent committee)

DVT/PE: 0.4% vs 0.2% (all patients had identifiable risk factors)

Malignancies: 0.6% vs 0.5% (consistent with RA background rates)

Laboratory: mean Hgb -0.3 g/dL; ANC stable; no Grade 4 cytopenias

Creatinine: mean +0.04 mg/dL from baseline, stable throughout

4. Discussion

XR-2041 15 mg once daily met its primary endpoint with an ACR20 response of 70.4% at Week 12, versus 29.1% for placebo ($p < 0.0001$). This effect size is consistent with or exceeds approved JAK inhibitors in similar RA populations.

The selectivity profile of XR-2041 (>200 -fold JAK1/3 over JAK2) was associated with a stable haematological profile: no Grade 3/4 anaemia, neutropenia, or thrombocytopenia events were observed. MACE and VTE rates were low and consistent with the expected background rate for an RA population of this age and CV profile.

Limitations include the 12-week primary endpoint (regulatory minimum) and the exclusion of patients with eGFR <30 , limiting generalisability to severe CKD.

5. Conclusion

XR-2041 15 mg once daily demonstrated statistically significant and clinically meaningful superiority over placebo on ACR20 at Week 12 and across all key secondary endpoints including DAS28-CRP remission, ACR50/70, HAQ-DI improvement, and radiographic non-progression at Week 52. The safety profile was acceptable with no new signals beyond the class effect of JAK inhibitors.

These results support XR-2041 as a potential new treatment option for patients with moderate-to-severe RA with inadequate response to methotrexate.